Degradation-Aware Solar + Battery + EV Charging Optimization Framework

Ahmedabad, India Case Study

Project Report

System: Rooftop PV + Battery Storage + Residential EV Charging

Tools: Python, pvlib, CVXPY, pandas, matplotlib, Streamlit

Weather Source: PVGIS Typical Meteorological Year (TMY) data

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1. Introduction

Residential electricity demand is increasing because of higher appliance usage, rooftop solar adoption, and electric vehicle charging. Solar PV can reduce grid dependency, but solar generation is not always available when the load is high. A battery can store excess solar energy and supply it later, but frequent charging and discharging also causes battery degradation.

This project develops a Python-based simulation and optimization framework for a residential solar PV, battery storage, and EV charging system. Ahmedabad, Gujarat, India is used as the case-study location. The system uses real Ahmedabad PVGIS Typical Meteorological Year weather data to estimate PV generation. The model also includes household load, EV charging demand, battery SOC tracking, battery SOH tracking, grid import/export calculation, controller comparison, long-term simulation, PV/battery sizing optimization, and a Streamlit dashboard.

2. Project Objective

The main engineering question is:

> *How should a residential solar + battery + EV charging system be controlled and sized so that it reduces electricity cost while also limiting battery degradation?*

The objectives are to model PV generation using Ahmedabad weather data, simulate residential and EV demand, implement battery dispatch, track battery health, compare controllers, and estimate long-term cost and degradation performance.

3. Case Study Location

Ahmedabad, Gujarat, India was selected because it has strong rooftop solar potential and is a relevant urban case for residential solar-plus-storage and EV charging applications. The model uses Asia/Kolkata local time. PVGIS timestamps are converted from UTC to local Ahmedabad time before simulation.

Item	Value
City	Ahmedabad, Gujarat, India
Timezone	Asia/Kolkata
Weather dataset	PVGIS Typical Meteorological Year (TMY)
Simulation step	1 hour
Base PV system	5 kW
Base battery	10 kWh

4. System Architecture

The modeled energy system contains rooftop solar PV, a residential load, an EV charging load, a stationary battery, and the utility grid. At every timestep, the simulator calculates PV generation, home load, EV charging demand, battery charge/discharge, grid import/export, energy cost, and battery degradation.

Main modules: PV model, load model, EV model, battery SOC model, degradation model, tariff model, controllers, optimizer, long-term simulator, sizing optimizer, plotting, and Streamlit dashboard.

5. PV Generation Model

The PV model uses PVGIS TMY weather data for Ahmedabad. Irradiance and temperature data are used with pvlib to compute solar position and plane-of-array irradiance. PV output is then estimated using system size and derating assumptions.

PV power = PV size x plane-of-array irradiance / 1000 x derating factor

Parameter	Assumption
PV system size	5 kW base case
Tilt angle	23 degrees
Azimuth	180 degrees
Derating factor	0.85
Weather variables	GHI, DNI, DHI, air temperature

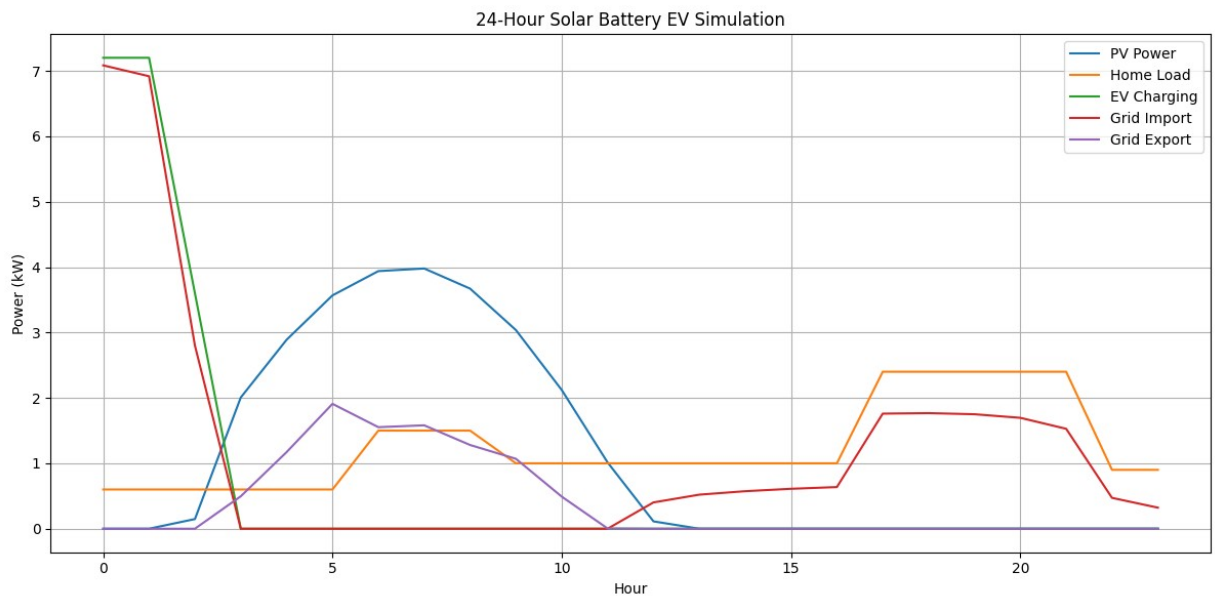

Figure 1. Example 24-hour power flow using Ahmedabad weather-based PV generation.

6. Home Load and EV Charging Model

The home load model uses a realistic synthetic residential profile. Load is lower overnight, moderate during daytime, and higher in the evening. The EV model represents a residential overnight charging case.

Model item	Assumption
EV arrival time	6 PM
EV departure time	8 AM
Daily EV energy requirement	18 kWh
EV charger power	7.2 kW
Load type	Residential daily profile

7. Battery SOC Model

The battery model tracks state of charge, efficiency, charge power, discharge power, and SOC operating limits. SOC is constrained between a safe minimum and maximum value to avoid unrealistic operation.

Battery parameter	Value
Capacity	10 kWh
Initial SOC	50%
Minimum SOC	20%
Maximum SOC	90%
Max charge power	5 kW
Max discharge power	5 kW
Efficiency	95%

8. Battery Degradation Model

Battery degradation is modeled using a simple control-oriented approach. Total capacity loss is represented as a combination of calendar aging and cycle aging. Calendar aging depends on time, while cycle aging depends on total battery throughput.

Capacity loss = calendar aging + cycle aging

The model tracks battery SOH, capacity loss percentage, cumulative battery throughput, and estimated usable capacity. This enables the optimization controller to include a battery wear cost rather than only minimizing grid energy cost.

9. Electricity Tariff Model

The tariff model calculates import cost and export revenue. The base tariff assumptions use Rs 8/kWh for grid import and Rs 3/kWh for solar export. Net energy cost is calculated as import cost minus export revenue.

10. Control Strategies

Five control strategies are compared: no battery, naive controller, rule-based controller, optimization controller, and degradation-aware optimization controller.

Controller	Description
No battery	PV directly serves load; deficit is imported and surplus is exported.
Naive	Charges battery with extra PV and discharges during deficits.
Rule-based	Uses simple peak-hour logic and avoids some unnecessary cycling.
Optimization	Minimizes grid import cost minus export revenue.
Degradation-aware	Minimizes energy cost plus battery degradation cost.

11. Optimization Formulation

The optimization controller is implemented with CVXPY. Decision variables include battery charge power, battery discharge power, grid import, grid export, and SOC trajectory. The core energy balance is:

PV + battery discharge + grid import = home load + EV charging + battery charge + grid export

The degradation-aware objective is:

minimize grid import cost - export revenue + battery degradation cost

12. Key 24-Hour Results

Metric	Value
Total grid import	28.84 kWh
Total grid export	9.55 kWh
Total energy cost	Rs 202.06
Total degradation cost	Rs 25.30
Total net cost	Rs 227.36
Final battery SOC	20.00%
Final battery SOH	99.9811%
Battery throughput	16.87 kWh

13. Controller Comparison

The controller comparison shows how different dispatch strategies affect grid import, cost, throughput, and final SOH. The no-battery case has the highest grid import and cost. Battery-based strategies reduce grid import. The degradation-aware controller includes wear cost, so its net cost includes both energy cost and battery aging cost.

Controller	Grid Import (kWh)	Net Cost (Rs)	Throughput (kWh)	Final SOH (%)
no_battery	38.34	255.95	0.00	100.0000
naive	28.84	202.06	16.87	99.9811
rule_based	31.69	215.39	10.86	99.9871
optimization	28.84	202.06	16.87	99.9811
degradation_aware	28.84	227.36	16.87	99.9811

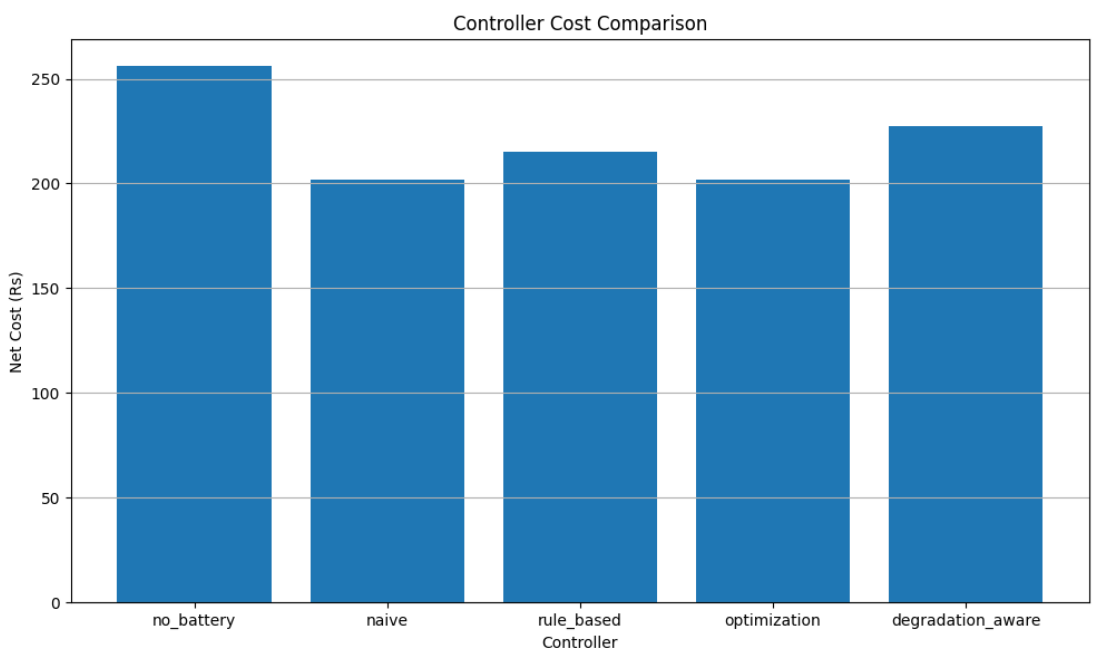

Figure 2. Controller net cost comparison.

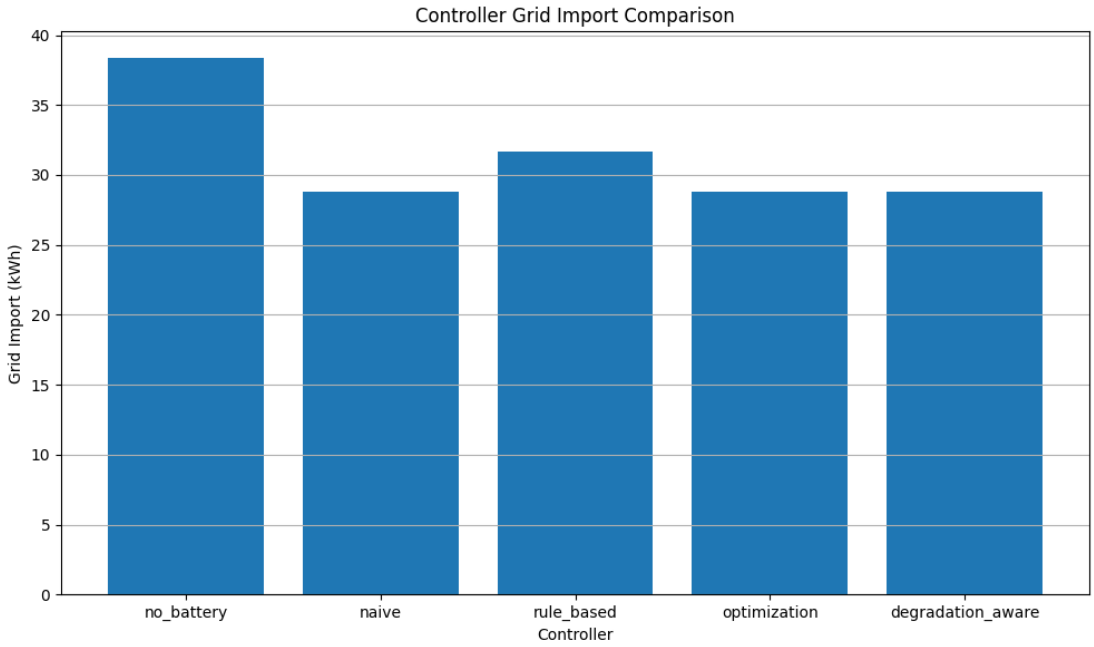

Figure 3. Controller grid import comparison.

14. 365-Day Long-Term Simulation

A full-year simulation was added using Ahmedabad PVGIS TMY weather data. The optimizer solves a 24-hour dispatch problem day by day. Battery SOC is carried from one day to the next, and SOH is tracked across the year.

Annual metric	Value
Annual grid import	11547.89 kWh
Annual grid export	2587.59 kWh
Annual energy cost	Rs 84620.36
Annual degradation cost	Rs 7008.06
Annual net cost	Rs 91628.42
Annual battery throughput	4672.04 kWh
Final SOH	94.60%
Estimated usable capacity	9.46 kWh

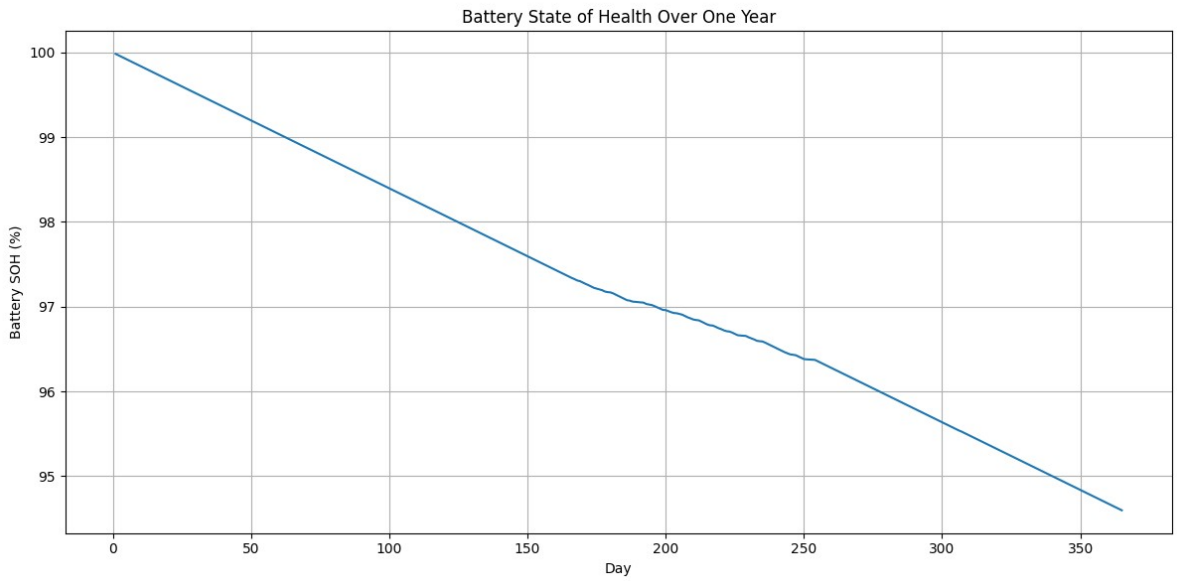

Figure 4. Battery SOH over one year.

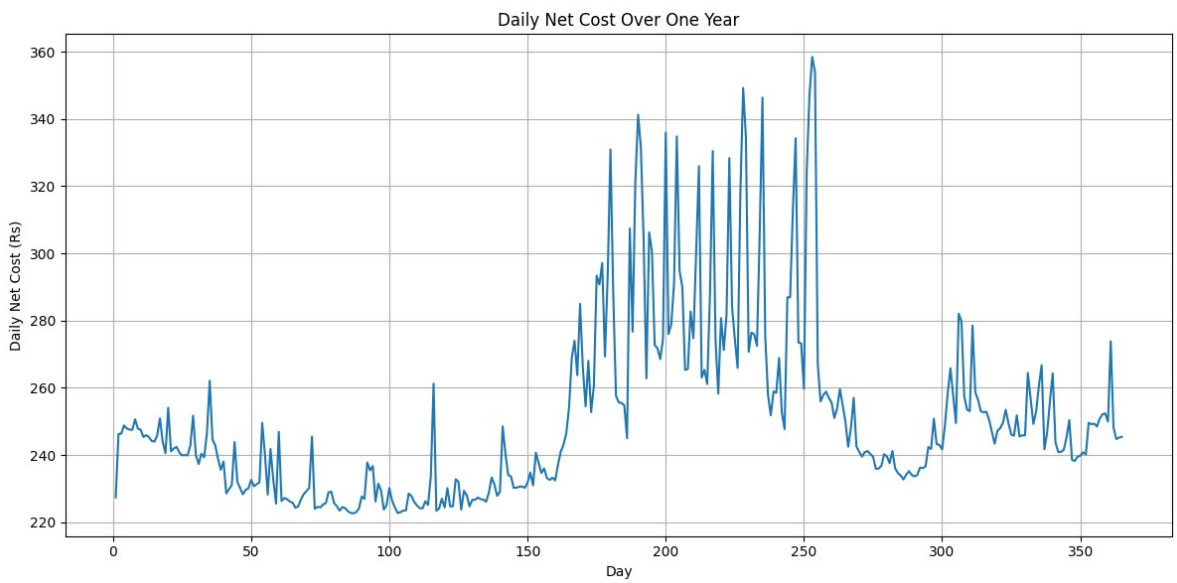

Figure 5. Daily net cost over one year.

15. PV and Battery Sizing Optimization

The sizing optimization tests PV sizes from 3 kW to 8 kW and battery sizes from 5 kWh to 15 kWh. Each design is evaluated using representative days from the Ahmedabad weather year and annualized to estimate yearly performance.

Best design metric	Value
PV size	8.0 kW
Battery size	15.0 kWh
Battery power limit	7.5 kW
Annual grid import	10817.32 kWh
Annual net cost	Rs 71893.34
Final SOH	99.81%
Estimated usable capacity	14.97 kWh

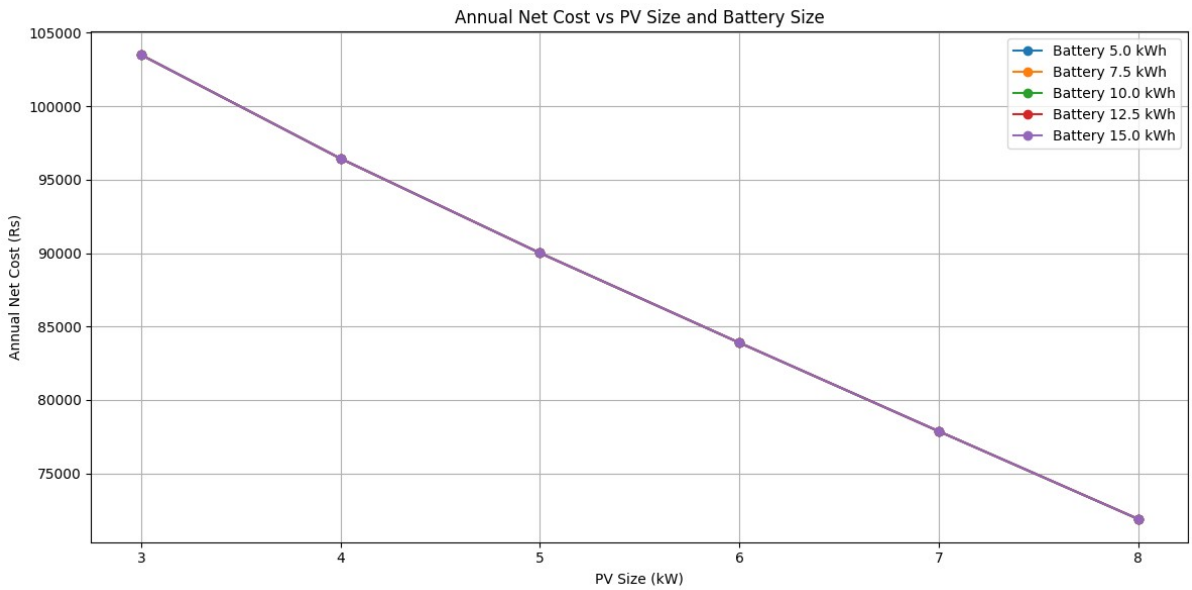

Figure 6. Annual net cost across PV and battery sizes.

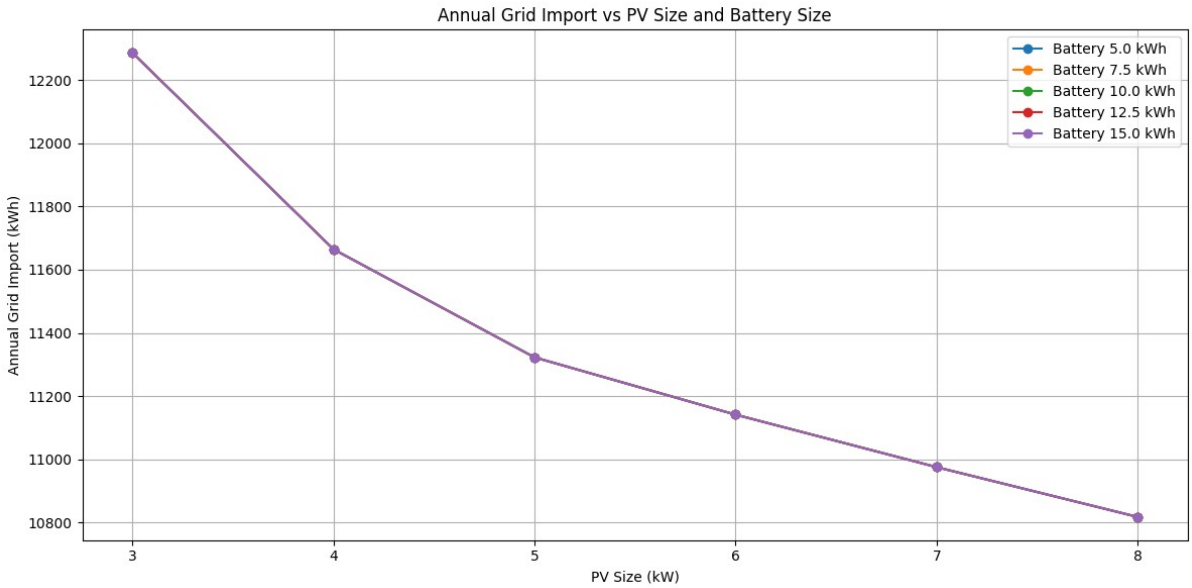

Figure 7. Annual grid import across PV and battery sizes.

16. Streamlit Dashboard

A Streamlit dashboard was created to make the project interactive. It includes pages for overview, 24-hour simulation, controller comparison, 365-day simulation, sizing optimization, and generated files. The dashboard is run with:

```
python3 -m streamlit run app.py
```

17. Results and Discussion

- Solar PV reduces grid import during daylight hours.
- Battery storage shifts solar energy to later demand periods.
- EV charging significantly increases residential load.
- Battery degradation is important because aggressive battery use increases throughput and reduces SOH.
- Degradation-aware optimization provides a more realistic strategy than cost-only optimization.
- Long-term simulation shows gradual SOH reduction over the year.
- Sizing optimization helps compare different PV and battery combinations.

18. Limitations

- Home load profile is synthetic.
- EV charging pattern is fixed.
- Battery degradation model is simplified and semi-empirical.
- Tariff model uses fixed import/export prices.
- PV sizing optimization can be improved by adding detailed capital cost modeling.
- Battery thermal behavior and inverter efficiency maps are not included.
- Forecast uncertainty is not included.

19. Future Improvements

- Use real residential load data.
- Add real Ahmedabad time-of-use tariff.
- Add EV arrival and departure randomness.
- Include PV and battery capital cost in more detail.
- Add battery replacement cost and payback period.
- Add multi-year degradation simulation.
- Include forecast uncertainty and battery temperature effects.
- Extend the model to multi-EV or fleet charging.

20. Conclusion

This project developed a degradation-aware solar PV, battery storage, and EV charging optimization framework for Ahmedabad, India. It uses real Ahmedabad PVGIS weather data for PV generation and combines it with residential load, EV charging, battery SOC tracking, battery SOH tracking, cost calculation, controller comparison, long-term simulation, sizing optimization, and a Streamlit dashboard.

The project shows that energy management should not only focus on reducing electricity cost, but should also consider battery degradation. By adding degradation cost into the optimization objective, the controller becomes more realistic and better suited for long-term residential energy management. Overall, the project demonstrates practical skills in renewable energy modeling, battery storage simulation, optimization, Python programming, data analysis, and dashboard development.